Mathematical Foundations of Deep Neural Networks

Lecture Notes, Part I: Foundations (Ch. 1–4)

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Chapter 1. Linear Regression and Gradient Descent

Regression, normal equations, gradient descent, generalization

1 The Regression Problem

In regression we observe pairs $(\mathbf{x}_i, y_i)$, $i = 1, \dots, n$, with input $\mathbf{x}_i \in \mathbb{R}^k$ and real response y_i, and we want a function that predicts y from $\mathbf{x}$. The simplest choice is a linear function. We motivate the choice of "best line" by the sum of squared errors (SSE): a horizontal benchmark line gives one SSE, rotating it changes the SSE, and plotting SSE against the rotation angle gives a curve that first decreases and then increases. The best line is the minimizer of this curve.

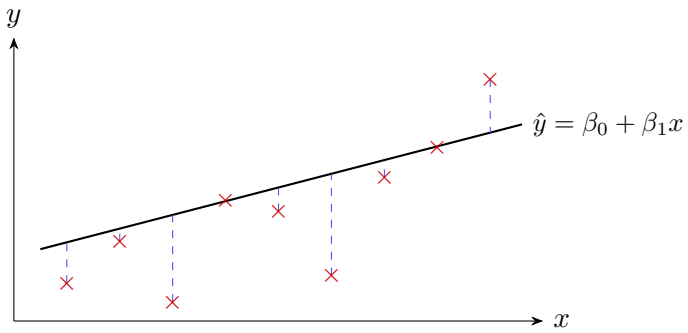
Definition 1.1: linear regression model and design matrix

$$y_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} + \varepsilon_i, \qquad i = 1, \dots, n.$$

Absorb the intercept with a constant input 1: let $d = k + 1$, $\boldsymbol{\beta} = (\beta_0, \dots, \beta_k)^\top \in \mathbb{R}^d$, and let the **design matrix** $X \in \mathbb{R}^{n \times d}$ have ith row $(1, x_{i1}, \dots, x_{ik})$. Then

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad f(\boldsymbol{\beta}) = \sum_{i=1}^{n} (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 = \|\mathbf{y} - X\boldsymbol{\beta}\|^2 = (\mathbf{y} - X\boldsymbol{\beta})^\top (\mathbf{y} - X\boldsymbol{\beta}).$$

Rows of X index samples; columns index features.



Residuals $y_i - \hat{y}_i$ (dashed) of a fitted line; least squares minimizes the sum of their squares.

2 Matrix Calculus Toolkit

Motivation. Every learning algorithm in the course minimizes a scalar loss over a vector or matrix of parameters. Rather than differentiate coordinate by coordinate, we collect a few identities for gradients of linear and quadratic forms that make such computations one-line.

We use the gradient convention: the gradient of a scalar with respect to a vector is a column vector of the same size.

Theorem 1.2: basic gradients

For $\mathbf{a} \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$,

$$\nabla_{\boldsymbol{\beta}}(\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}, \qquad \nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = (A + A^\top)\boldsymbol{\beta}, \qquad \nabla^2_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\top A \boldsymbol{\beta}) = A + A^\top.$$

If A is symmetric the last two become $2A\boldsymbol{\beta}$ and $2A$.

Proof. $\mathbf{a}^\top\boldsymbol{\beta} = \sum_j a_j\beta_j$, so $\partial/\partial\beta_k = a_k$. For the quadratic form, $\boldsymbol{\beta}^\top A\boldsymbol{\beta} = \sum_{i,j} A_{ij}\beta_i\beta_j$. Differentiating with respect to β_k, the product rule picks up the terms with $i = k$ and those with $j = k$:

$$\frac{\partial}{\partial\beta_k}\sum_{i,j} A_{ij}\beta_i\beta_j = \sum_j A_{kj}\beta_j + \sum_i A_{ik}\beta_i = (A\boldsymbol{\beta})_k + (A^\top\boldsymbol{\beta})_k.$$

Differentiating once more gives the constant matrix $A + A^\top$. ■

3 Normal Equations

Motivation. Least squares is the one learning problem with a closed-form solution, and it is the benchmark for everything else. Setting the gradient to zero gives the normal equations; their geometry, projection of $\mathbf{y}$ onto the column space, explains what the fit is doing.

Theorem 1.3: least squares solution

$\nabla f(\boldsymbol{\beta}) = 2X^\top(X\boldsymbol{\beta} - \mathbf{y})$ and $\nabla^2 f = 2X^\top X \succeq 0$. Every minimizer solves the **normal equations**

$$X^\top X\boldsymbol{\beta} = X^\top\mathbf{y}.$$

If X has full column rank d, then $X^\top X \succ 0$ and the unique minimizer is $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top\mathbf{y}$.

Proof. Expand $f = \mathbf{y}^\top\mathbf{y} - 2\mathbf{y}^\top X\boldsymbol{\beta} + \boldsymbol{\beta}^\top X^\top X\boldsymbol{\beta}$ (the two cross terms are equal scalars). With the toolkit, $\nabla f = -2X^\top\mathbf{y} + 2X^\top X\boldsymbol{\beta}$ and $\nabla^2 f = 2X^\top X$. For every $\mathbf{v}$, $\mathbf{v}^\top X^\top X\mathbf{v} = \|X\mathbf{v}\|^2 \geq 0$, so f is convex and stationary points are global minimizers. If the columns of X are linearly independent, $X\mathbf{v} \neq \mathbf{0}$ whenever $\mathbf{v} \neq \mathbf{0}$; then $X^\top X \succ 0$, it is invertible, f is strictly convex and the minimizer is unique. ■

Remark 1.4: geometry and rank

The normal equations say $X^\top(\mathbf{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$: the residual is orthogonal to every column of X, so $X\hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space. Since the first column is $\mathbf{1}_n$, the residuals sum to zero. Having $n > d$ does not by itself guarantee full rank; with dependent columns the minimizer is not unique and the pseudoinverse gives the minimum-norm solution. Numerically one solves the linear system by QR or SVD rather than forming an inverse.

Example 1.5: least squares on three points

Points $(0, 1), (1, 2), (2, 2)$ and model $y = \beta_0 + \beta_1 x$:

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad X^\top X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \quad X^\top\mathbf{y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, \quad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 5 & -3 \\ -3 & 3 \end{pmatrix}.$$

Hence $\hat{\boldsymbol{\beta}} = \frac{1}{6}(25 - 18, -15 + 18)^\top = (7/6,\ 1/2)^\top$. Predictions are $(7/6, 5/3, 13/6)$, residuals $(-1/6, 1/3, -1/6)$ (sum zero), and SSE $= 1/6$.

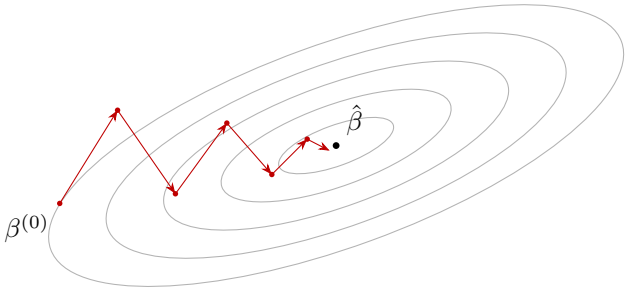
4 Gradient Descent

4.1 The algorithm

Instead of solving the normal equations directly we may iterate

$$\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta \nabla f(\boldsymbol{\beta}_t) = \boldsymbol{\beta}_t + 2\eta X^\top (\mathbf{y} - X\boldsymbol{\beta}_t),$$

i.e. componentwise $\beta_j \leftarrow \beta_j + 2\eta \sum_i (y_i - \hat{y}_i) x_{ij}$ with $x_{i0} = 1$. The factor 2 is often absorbed into the learning rate η. Each full-batch step uses all samples. The negative gradient is the direction of steepest descent: for a unit vector $\mathbf{v}$ the first-order change $\nabla f^\top \mathbf{v}$ is minimized, by Cauchy–Schwarz, at $\mathbf{v} = -\nabla f / \|\nabla f\|$.



Gradient descent on the level sets of a quadratic objective: each step moves against the gradient, so an elongated bowl produces a zigzag path.

4.2 Convergence on least squares

Write $J(\boldsymbol{\beta}) = \frac{1}{2}\|X\boldsymbol{\beta} - \mathbf{y}\|^2$ (so $\nabla J = X^\top(X\boldsymbol{\beta} - \mathbf{y})$) and $H = X^\top X$, assumed positive definite.

Theorem 1.6: exact convergence condition

Gradient descent $\boldsymbol{\beta}_{t+1} = \boldsymbol{\beta}_t - \eta \nabla J(\boldsymbol{\beta}_t)$ converges to $\hat{\boldsymbol{\beta}}$ for every initial point if and only if

$$0 < \eta < \frac{2}{\lambda_{\max}(X^\top X)}.$$

Proof. Error recursion. Since $\nabla J(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, $X^\top \mathbf{y} = H\hat{\boldsymbol{\beta}}$, so $\nabla J(\boldsymbol{\beta}_t) = H(\boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}})$. With $\mathbf{e}_t = \boldsymbol{\beta}_t - \hat{\boldsymbol{\beta}}$,

$$\mathbf{e}_{t+1} = \mathbf{e}_t - \eta H \mathbf{e}_t = (\mathbf{I} - \eta H)\mathbf{e}_t, \qquad \mathbf{e}_t = (\mathbf{I} - \eta H)^t \mathbf{e}_0.$$

Diagonalize. H is symmetric, so $H = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$. Using $\mathbf{I} = QQ^\top$, $\mathbf{I} - \eta H = Q(\mathbf{I} - \eta\Lambda)Q^\top$. Let $\mathbf{c}_t = Q^\top \mathbf{e}_t$. Multiplying the recursion by $Q^\top$ and using $Q^\top Q = \mathbf{I}$ gives $\mathbf{c}_{t+1} = (\mathbf{I} - \eta\Lambda)\mathbf{c}_t$, a diagonal recursion:

$$c_{t,i} = (1 - \eta\lambda_i)^t \, c_{0,i}.$$

Because $\|\mathbf{c}_t\| = \|\mathbf{e}_t\|$, $\mathbf{e}_t \to \mathbf{0}$ iff every coordinate $c_{t,i} \to 0$.

Sufficiency. If $|1 - \eta\lambda_i| < 1$ for all i, every coordinate tends to zero for every $\mathbf{e}_0$.

Necessity. If $|1 - \eta\lambda_i| \geq 1$ for some i, start from $\mathbf{e}_0 = \mathbf{q}_i$ (the ith eigenvector). Then $\mathbf{e}_t = (1 - \eta\lambda_i)^t \mathbf{q}_i$ and $\|\mathbf{e}_t\| = |1 - \eta\lambda_i|^t \geq 1$, so this initial point does not converge.

Solve. $|1 - \eta\lambda_i| < 1 \iff 0 < \eta\lambda_i < 2 \iff 0 < \eta < 2/\lambda_i$. Requiring this for all i is equivalent to $0 < \eta < 2/\lambda_{\max}$. ∎

Remark 1.7: conditioning

The contraction factor of direction i is $|1-\eta\lambda_i|$. The rate $\max_i |1-\eta\lambda_i|$ is minimized by $\eta^\star = 2/(\lambda_{\max}+\lambda_{\min})$, giving rate $(\kappa-1)/(\kappa+1)$ with condition number $\kappa = \lambda_{\max}/\lambda_{\min}$. Elongated (ill-conditioned) level sets make gradient descent slow; this is revisited in Chapter 6. At $\eta = 2/\lambda_{\max}$ the steepest direction oscillates with factor -1 and never decays, so the inequality is strict. In Example 1.5, $\lambda(X^\top X) = 4 \pm \sqrt{10}$ and the condition is $0 < \eta < (4-\sqrt{10})/3 \approx 0.279$.

5 Polynomial Features and Generalization

5.1 Feature maps

For scalar input take $\boldsymbol{\phi}_M(x) = (1, x, x^2, \dots, x^M)^\top$ and $y(x) = \sum_{j=0}^{M} \beta_j x^j$. The model is nonlinear in x but linear in the coefficients, so all of the above applies with the (Vandermonde-type) design matrix whose rows are $\boldsymbol{\phi}_M(x_i)^\top$.

5.2 Training error versus generalization

Because a degree-M model contains every lower-degree model, the minimal training SSE cannot increase as M increases. Validation error can nevertheless increase: a high-degree fit follows noise and oscillates between data points (overfitting), while too low a degree cannot capture the trend (underfitting). Performance is measured by the root mean squared error

$$\mathrm{MSE} = \frac{1}{n}\mathrm{SSE}, \qquad E_{\mathrm{RMS}} = \sqrt{\mathrm{MSE}}.$$

Remark 1.8: roles of the data splits

Training data choose the coefficients; validation data choose the degree and the regularization strength; a held-out test set estimates final performance. In typical learning curves the gap between training and validation RMS grows with M for $M \geq 5$, and degree 9 with $n = 10$ overfits badly while $n = 100$ points support the same degree much better. "About ten points per parameter" is a heuristic, not a theorem. Regularization (Chapter 3) is the second remedy.

6 Proof Supplement: Uniqueness and the Pseudoinverse

Motivation. When $X^\top X$ is singular the normal equations have many solutions. The pseudoinverse picks the one of minimum norm, which is also the solution gradient descent converges to from zero, an early example of implicit regularization.

Theorem 1.9: set of minimizers

The minimizers of $\|\mathbf{y} - X\boldsymbol{\beta}\|^2$ are exactly the solutions of $X^\top X\boldsymbol{\beta} = X^\top \mathbf{y}$; this system is always consistent, and the solution set is $\hat{\boldsymbol{\beta}}_0 + \ker X$ for any particular solution $\hat{\boldsymbol{\beta}}_0$. Among them, $X^+\mathbf{y}$ has minimal Euclidean norm.

Proof. Consistency: $X^\top \mathbf{y}$ lies in the column space of $X^\top$, which equals the column space of $X^\top X$ (both have orthogonal complement $\ker X$, since $X^\top X\mathbf{v} = \mathbf{0} \Rightarrow \|X\mathbf{v}\|^2 = 0$). Two solutions differ by $\mathbf{v}$ with $X^\top X\mathbf{v} = \mathbf{0}$, i.e. $\mathbf{v} \in \ker X$, and give the same fitted values. Write $X = U\Sigma V^\top$ (thin

SVD with r positive singular values). Every solution is $V_r \Sigma_r^{-1} U_r^\top \mathbf{y} + \mathbf{v}$ with $\mathbf{v} \perp \text{range}(V_r)$; by Pythagoras its norm is minimized at $\mathbf{v} = \mathbf{0}$, which is $X^+ \mathbf{y}$. ■

Chapter 2. Probability, Likelihood and Information Theory

Random variables, maximum likelihood, MAP estimation, Bayesian models; Entropy, KL divergence, mutual information, cross-entropy

1 Events and Conditional Probability

A sample space Ω contains the possible outcomes and events are (measurable) subsets. The axioms are nonnegativity, $P(\Omega) = 1$ and countable additivity for disjoint events. They imply

$$P(\emptyset) = 0, \qquad P(E^c) = 1 - P(E), \qquad P(E \cup F) = P(E) + P(F) - P(E \cap F),$$

monotonicity $E \subseteq F \Rightarrow P(E) \leq P(F)$, and continuity along increasing or decreasing sequences of events.

Definition 2.1: conditional probability, Bayes' rule, independence

For $P(E) > 0$, $P(F \mid E) = P(F \cap E)/P(E)$. For a partition $E_1, E_2, \ldots$ of Ω,

$$P(F) = \sum_i P(F \mid E_i)P(E_i), \qquad P(E_i \mid F) = \frac{P(F \mid E_i)P(E_i)}{\sum_j P(F \mid E_j)P(E_j)}.$$

E, F are independent if $P(E \cap F) = P(E)P(F)$.

Example 2.2: why the prior matters

Let $P(D) = 0.01$, $P(+ \mid D) = 0.9$, $P(+ \mid D^c) = 0.1$. Then

$$P(D \mid +) = \frac{0.9(0.01)}{0.9(0.01) + 0.1(0.99)} \approx 0.0833.$$

A sensitive test does not make the posterior large when the prior is small. Two fair coin tosses: "first head" and "second head" each have probability 1/2 and their intersection 1/4, so they are independent.

2 Random Variables and Distributions

Motivation. Learning treats data as samples from an unknown distribution, and losses are expectations. We fix the notation for distributions, expectations and independence used in every later chapter.

A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (e.g. the number of heads in three tosses, $X(HHT) = 2$). Its CDF is $F_X(t) = P(X \leq t)$.

- Discrete: probability mass $p(x) = P(X = x)$, $F_X(t) = \sum_{x \leq t} p(x)$.
- Continuous: density $f \geq 0$ with $\int f = 1$, $P(a < X \leq b) = \int_a^b f(u)\,du$. A density value is not a probability and may exceed one; $P(X = x) = 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \text{ or } \int g(x)f(x)\,dx, \qquad \mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Theorem 2.3: tools used throughout the course

(i) Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$ always.

(ii) Independence: $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$; hence $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. Uncorrelatedness ($\operatorname{Cov} = 0$) suffices for the variance identity.

(iii) Tower property: $\mathbb{E}[\mathbb{E}[Z \mid X]] = \mathbb{E}Z$.

(iv) $\mathbb{E}[(Z-c)^2] = \operatorname{Var}Z + (\mathbb{E}Z - c)^2 \geq \operatorname{Var}Z$ for every constant c.

Proof. (iv): write $Z - c = (Z - \mathbb{E}Z) + (\mathbb{E}Z - c)$, square, and use $\mathbb{E}[Z - \mathbb{E}Z] = 0$ to kill the cross term. The other items are standard. ■

3 Maximum Likelihood Estimation

Definition 2.4: likelihood and MLE

For iid data $x_1, \ldots, x_n$ from $p(x \mid \theta)$,

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta), \qquad \hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

The data are fixed and the parameter varies; L is not a distribution over θ. The logarithm preserves the maximizer and turns products into sums.

Example 2.5: Bernoulli, including the boundary

$x_i \in \{0, 1\}$ with $P(x_i = 1) = \theta$; let $s = \sum x_i$, $f = n - s$. Then $\log L = s \log\theta + f \log(1-\theta)$. For $0 < s < n$,

$$\frac{d}{d\theta} \log L = \frac{s}{\theta} - \frac{f}{1-\theta} = 0 \iff \theta = \frac{s}{n}, \qquad \frac{d^2}{d\theta^2} \log L < 0.$$

If $s = 0$, $L = (1-\theta)^n$ is decreasing and the maximizer is 0; if $s = n$ it is 1. Both agree with s/n, but the derivative argument cannot be used there. Eight heads in ten tosses gives 0.8; two positive reviews out of two gives the extreme estimate 1.

Example 2.6: Gaussian

For $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, $\log L = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$, giving $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$ (biased; the unbiased version divides by $n - 1$).

4 Bayesian Inference and MAP

Motivation. Maximum likelihood can overfit with little data. Bayesian inference adds a prior and returns a posterior; its mode (MAP) is a single estimate that will turn out to be maximum likelihood plus a regularizer.

Definition 2.7: prior, posterior, evidence, MAP

With prior $p(\theta)$ and likelihood $p(D \mid \theta)$,

$$p(\theta \mid D) = \frac{p(D \mid \theta)p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta)p(\theta)\, d\theta,$$

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \{\log p(D \mid \theta) + \log p(\theta)\}.$$

The evidence does not depend on θ. MAP is the posterior mode; the posterior mean is a different summary.

4.1 Beta–Bernoulli conjugacy

Theorem 2.8: Beta prior

If $\theta \sim \mathrm{Beta}(a, b)$, i.e. $p(\theta) = \theta^{a-1}(1-\theta)^{b-1}/B(a,b)$ with $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and s successes, f failures are observed, then

$$\theta \mid D \sim \mathrm{Beta}(a+s,\, b+f), \qquad \mathbb{E}[\theta \mid D] = \frac{a+s}{a+b+n}, \qquad \hat{\theta}_{\mathrm{MAP}} = \frac{a+s-1}{a+b+n-2}$$

(the mode formula needs both posterior parameters above one).

Proof. $p(\theta \mid D) \propto \theta^s(1-\theta)^f \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+s)-1}(1-\theta)^{(b+f)-1}$, which is the $\mathrm{Beta}(a+s, b+f)$ kernel. For $\mathrm{Beta}(\alpha,\beta)$, $\mathbb{E}\theta = B(\alpha+1,\beta)/B(\alpha,\beta) = \alpha/(\alpha+\beta)$ using $\Gamma(z+1) = z\Gamma(z)$. The mode solves $\frac{\alpha-1}{\theta} = \frac{\beta-1}{1-\theta}$, i.e. $\theta = \frac{\alpha-1}{\alpha+\beta-2}$. ∎

The uniform prior is $\mathrm{Beta}(1,1)$; then the posterior mean $(s+1)/(n+2)$ is Laplace's rule of succession, and it equals the posterior predictive probability of a success, since $P(X_{\mathrm{new}} = 1 \mid D) = \int \theta\, p(\theta \mid D)\, d\theta$.

Example 2.9: the two sellers

Seller 1 has 90 positive and 10 negative reviews; seller 2 has 2 positive and 0 negative. With uniform priors

$$\theta_1 \mid D_1 \sim \mathrm{Beta}(91, 11), \qquad \theta_2 \mid D_2 \sim \mathrm{Beta}(3, 1).$$

Posterior predictive probabilities of a positive next review: $91/102 \approx 0.892$ versus $3/4 = 0.75$, so seller 1 is preferred, whereas the MLE (0.9 versus 1) prefers seller 2. A different question is which unknown rate is larger: since $\mathrm{Beta}(3,1)$ has CDF t^3,

$$P(\theta_1 > \theta_2 \mid D) = \mathbb{E}[\theta_1^3] = \frac{91 \cdot 92 \cdot 93}{102 \cdot 103 \cdot 104} \approx 0.713.$$

5 MAP as Regularization and the Bias–Variance Decomposition

Motivation. A Gaussian prior turns MAP into ℓ_2-penalized likelihood, so regularization has a probabilistic meaning. The bias–variance decomposition then explains the trade-off: shrinking adds bias but can remove more variance.

5.1 Gaussian mean with a Gaussian prior

Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$ and prior $\theta \sim N(0, \tau^2)$.

Theorem 2.10: MAP is ridge, and it shrinks

$$\hat{\theta}_{\text{MAP}} = \arg\min_{\theta} \sum_{i=1}^{n} (X_i - \theta)^2 + \lambda\theta^2, \quad \lambda = \frac{1}{\tau^2}, \qquad \hat{\theta}_{\text{MAP}} = a\bar{X}, \quad a = \frac{n}{n+\lambda} = \frac{n\tau^2}{n\tau^2 + 1}.$$

Proof. $-\log p(\theta \mid X) = \frac{1}{2}\sum(X_i - \theta)^2 + \frac{\theta^2}{2\tau^2} + \text{const}$; multiplying by 2 gives the ridge objective. Setting the derivative $-2\sum(X_i - \theta) + 2\lambda\theta$ to zero gives $(n + \lambda)\theta = n\bar{X}$. ■

5.2 Bias–variance decomposition

Theorem 2.11: MSE = bias^2 + variance

For any estimator $\hat{\theta}$ of θ, $\mathbb{E}[(\hat{\theta} - \theta)^2] = (\mathbb{E}\hat{\theta} - \theta)^2 + \text{Var}(\hat{\theta})$.

Proof. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + (\mathbb{E}\hat{\theta} - \theta)$. The second bracket is a constant and the first has mean zero, so the cross term vanishes after taking expectations. ■

For $\hat{\theta} = a\bar{X}$: bias $(a - 1)\theta$, variance a^2/n, so $\text{MSE}(a) = (a - 1)^2\theta^2 + a^2/n$. Minimizing over a gives $a^\star = n\theta^2/(n\theta^2 + 1)$, which equals the MAP shrinkage factor exactly when $\tau^2 = \theta^2$. Since $a^\star < 1$, some bias always pays off in MSE relative to the MLE $a = 1$; this is the first instance of the bias–variance trade-off that reappears for ridge regression (Chapter 3) and Adam's bias correction (Chapter 6).

Remark 2.12: Laplace prior gives the lasso

With $p(\theta) \propto e^{-|\theta|/b}$ instead, $-\log$ posterior is $\frac{1}{2}\sum(X_i - \theta)^2 + \frac{1}{b}|\theta|$ and the MAP is the soft threshold $\text{sign}(\bar{X})\max(|\bar{X}| - \frac{1}{nb}, 0)$ (Chapter 3).

6 Information and Entropy

Motivation. Classification losses are built from information theory. Entropy measures the average surprise of a distribution, and it is the baseline from which cross-entropy and KL divergence measure the cost of using the wrong distribution.

The information of an outcome is $h(x) = -\log p(x)$: rare outcomes carry more information, and for independent events probabilities multiply while information adds.

Definition 2.13: entropy

For a discrete random variable with pmf p,

$$H(X) = -\sum_{x} p(x) \log p(x) = \mathbb{E}[-\log p(X)], \qquad 0 \log 0 := 0.$$

Natural logs give nats; base 2 gives bits.

Theorem 2.14: range of entropy

If X takes at most K values, $0 \le H(X) \le \log K$. $H = 0$ iff X is deterministic; $H = \log K$ iff X is uniform.

Proof. Each term $-p \log p \ge 0$ for $p \in [0,1]$, with equality iff $p \in \{0,1\}$. For the upper bound apply Jensen to the concave log: $H = \mathbb{E} \log \frac{1}{p(X)} \le \log \mathbb{E} \frac{1}{p(X)} = \log \sum_{x : p(x)>0} 1 \le \log K$, with equality iff $1/p(X)$ is constant on the support and the support has K points. ■

(For continuous densities the differential entropy can be negative; the statement above is for discrete variables.)

Definition 2.15: joint and conditional entropy

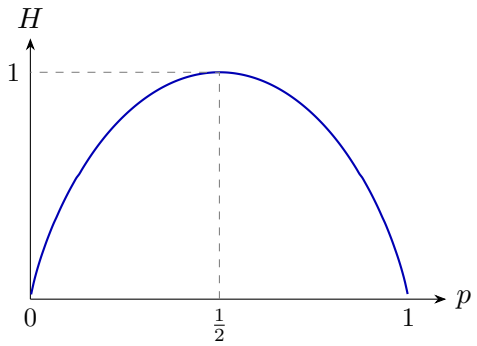
$$H(X,Y) = -\sum_{x,y} p(x,y) \log p(x,y),$$

$$H(Y \mid X) = -\sum_{x,y} p(x,y) \log p(y \mid x) = \sum_{x} p(x) H(Y \mid X = x).$$

Theorem 2.16: chain rule

$H(X,Y) = H(X) + H(Y \mid X)$.

Proof. Take $-\mathbb{E}$ of $\log p(X,Y) = \log p(X) + \log p(Y \mid X)$. ■



Entropy (in bits) of a Bernoulli(p) variable: zero for a certain outcome, maximal at $p = \frac{1}{2}$.

7 Kullback–Leibler Divergence

Motivation. How different are two distributions? KL divergence is the expected extra log-loss from coding with q instead of p. It is nonnegative and zero only at $q = p$, which is why minimizing it fits a model to data.

Definition 2.17: KL divergence

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x} p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_{p}\Big[\log \frac{p(X)}{q(X)}\Big].$$

Terms with $p(x) = 0$ contribute 0; if $p(x) > 0 = q(x)$ for some x the divergence is $+\infty$.

Theorem 2.18: Gibbs' inequality

$D_{\mathrm{KL}}(p \,\|\, q) \geq 0$, with equality iff $p = q$.

Proof. Sum over $S = \{x : p(x) > 0\}$. By Jensen for the concave log,

$$-D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x \in S} p(x) \log \frac{q(x)}{p(x)} \leq \log \sum_{x \in S} p(x) \frac{q(x)}{p(x)} = \log \sum_{x \in S} q(x) \leq \log 1 = 0.$$

Equality forces (i) $q(x)/p(x) = c$ constant on S (Jensen) and (ii) $\sum_S q = 1$; then $c = \sum_S q = 1$, so $q = p$. Conversely $p = q$ gives $\log 1 = 0$ term by term. ■

Example 2.19: KL is not symmetric

For Bernoulli parameters, $D(r \,\|\, s) = r \log \frac{r}{s} + (1-r) \log \frac{1-r}{1-s}$. With $r = 0.9$, $s = 0.5$: $D(r\|s) \approx 0.368$ but $D(s\|r) \approx 0.511$ nats. KL is not a metric (no symmetry, no triangle inequality).

8 Mutual Information

Motivation. Mutual information is the KL divergence between a joint distribution and the product of its marginals: how much knowing X reduces uncertainty about Y. It is zero exactly under independence.

Definition 2.20: mutual information

$$I(X;Y) = D_{\mathrm{KL}}(p_{XY} \,\|\, p_X p_Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

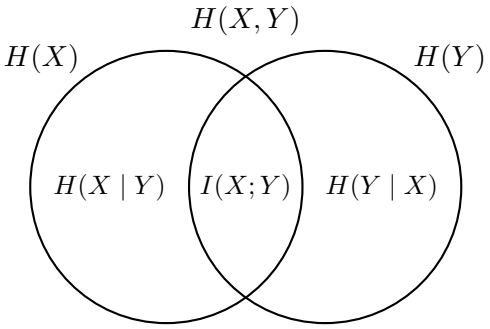
Theorem 2.21: identities

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y) \;\geq 0,$$

with $I = 0$ iff $X \perp Y$. Consequently $H(X \mid Y) \leq H(X)$ (conditioning reduces entropy).

Proof. $\log \frac{p(x,y)}{p(x)p(y)} = \log p(x \mid y) - \log p(x)$; take $\mathbb{E}$ and use $\sum_y p(x,y) = p(x)$ to get $-H(X \mid Y) + H(X)$. The last form follows from the chain rule. Nonnegativity and the equality case are Gibbs' inequality. ■

The entropy (Venn) diagram encodes $H(X,Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X)$. For a fair bit X and $Y = X$, $I = 1$ bit; for independent fair bits, $I = 0$.



Entropy, joint and conditional entropy, and mutual information: the union of the two disks is $H(X,Y)$ and the overlap is $I(X;Y)$.

9 Cross-Entropy

Motivation. Cross-entropy is entropy plus KL divergence, so minimizing it over the model is the same as minimizing KL. This is why the standard classification loss is cross-entropy, and why it equals negative log-likelihood.

Definition 2.22: cross-entropy

$H(p,q) = -\sum_x p(x)\log q(x) = \mathbb{E}_p[-\log q(X)]$.

Theorem 2.23: cross-entropy = entropy + KL

$H(p,q) = H(p) + D_{\mathrm{KL}}(p\,\|\,q)$. Hence for fixed p, $\arg\min_q H(p,q) = \arg\min_q D_{\mathrm{KL}}(p\,\|\,q) = p$.

Proof. $-\sum p\log q = -\sum p\log p + \sum p\log(p/q)$. The second statement uses Gibbs. ■

Remark 2.24: classification loss and soft labels

For a one-hot label $\mathbf{e}_y$, $H(\mathbf{e}_y) = 0$ and $H(\mathbf{e}_y, q) = D_{\mathrm{KL}}(\mathbf{e}_y\|q) = -\log q_y$: the classification loss. $H(p,q)$ is *linear in its first argument*, $H(\alpha p + \beta p', q) = \alpha H(p,q) + \beta H(p',q)$, because p enters only as coefficients. Therefore a soft label that averages n one-hot labels gives $H(\frac{1}{n}\sum_j \mathbf{e}_{y_j}, q) = \frac{1}{n}\sum_j(-\log q_{y_j})$: fitting the label distribution by KL is equivalent (up to the constant $H(\hat{p})$) to ordinary cross-entropy on every individual label.

10 Jensen–Shannon Divergence

Motivation. KL divergence is asymmetric and can be infinite. The Jensen–Shannon divergence symmetrizes it through the mixture $(p+q)/2$ and stays bounded; it is the divergence that the original GAN objective minimizes.

Definition 2.25: JS divergence

With $m = \frac{1}{2}(p+q)$, $D_{\mathrm{JS}}(p\,\|\,q) = \frac{1}{2}D_{\mathrm{KL}}(p\,\|\,m) + \frac{1}{2}D_{\mathrm{KL}}(q\,\|\,m)$.

Theorem 2.26: properties

(i) $D_{\mathrm{JS}}(p\|q) \geq 0$, with equality iff $p = q$.

(ii) $D_{\mathrm{JS}}(p\|q) = D_{\mathrm{JS}}(q\|p)$.

(iii) $D_{\mathrm{JS}}(p\|q) = H(m) - \frac{1}{2}H(p) - \frac{1}{2}H(q)$.

(iv) $D_{\mathrm{JS}}(p\|q) \leq \log 2$, with equality iff p and q have disjoint supports.

Proof. (i) Both KL terms are nonnegative by Gibbs, so $D_{\mathrm{JS}} \geq 0$. If $D_{\mathrm{JS}} = 0$ then $D_{\mathrm{KL}}(p\|m) = 0$, so $p = m = \frac{1}{2}(p+q)$, i.e. $p = q$; conversely $p = q$ gives $m = p$ and both terms vanish. Note that $D_{\mathrm{KL}}(p\|m) < \infty$ always, since $m \geq p/2$.

(ii) m is symmetric in p, q and the definition is a symmetric sum.

(iii) $D_{\mathrm{KL}}(p\|m) = -H(p) - \sum p \log m$; averaging with the q term gives $-\frac{1}{2}H(p) - \frac{1}{2}H(q) - \sum m \log m$.

(iv) Substituting m,

$$D_{\mathrm{JS}} = \log 2 + \tfrac{1}{2}\sum p \log \frac{p}{p+q} + \tfrac{1}{2}\sum q \log \frac{q}{p+q} \leq \log 2,$$

since $p/(p+q) \leq 1$ and $q/(p+q) \leq 1$. Equality iff $q = 0$ wherever $p > 0$ and vice versa. ■

Remark 2.27: JS as mutual information

Let B be a fair bit and $Z \sim p$ if $B = 0$, $Z \sim q$ if $B = 1$. Then $Z \sim m$, $H(Z \mid B) = \frac{1}{2}H(p) + \frac{1}{2}H(q)$, and by (iii) $I(Z; B) = D_{\mathrm{JS}}(p\|q) \leq H(B) = \log 2$.

Chapter 3. Probabilistic Regression, Regularization and Kernel Ridge

Linear regression from a probabilistic view, regularization, ridge regression, the kernel trick

1 Gaussian Likelihood Explains Squared Error

Motivation. Why squared error? If the noise is Gaussian, maximizing the likelihood is exactly minimizing the sum of squares. Each loss in the course corresponds in the same way to a noise model.

Theorem 3.1: MLE under Gaussian noise is least squares

If $y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, then

$$\log p(\mathbf{y} \mid X, \boldsymbol{\beta}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2.$$

For fixed σ^2, maximizing the likelihood is minimizing the SSE, so $\hat{\boldsymbol{\beta}}_{\text{MLE}}$ solves the normal equations. Maximizing over σ^2 then gives $\hat{\sigma}^2 = \text{SSE}/n$.

Proof. The density of y_i is $(2\pi\sigma^2)^{-1/2}\exp\{-(y_i - \mathbf{x}_i^\top\boldsymbol{\beta})^2/(2\sigma^2)\}$; independence multiplies densities and the log turns the product into the displayed sum. Differentiating in σ^2: $-\frac{n}{2\sigma^2} + \frac{\text{SSE}}{2\sigma^4} = 0$. ■

2 Ridge Regression

Motivation. Least squares is unstable when $X^\top X$ is nearly singular or there are more features than samples. Adding $\lambda\|\mathbf{w}\|^2$ makes the problem well posed, shrinks the poorly determined directions most, and trades a little bias for a large reduction in variance.

2.1 Definition and closed form

Definition 3.2: ridge

For $\lambda > 0$, $J(\boldsymbol{\beta}) = \frac{1}{2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|^2$.

Theorem 3.3: ridge solution

$\nabla J = X^\top X\boldsymbol{\beta} - X^\top\mathbf{y} + \lambda\boldsymbol{\beta}$ and the unique minimizer is $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda\mathbf{I})^{-1}X^\top\mathbf{y}$, even when X is rank deficient.

Proof. $\mathbf{v}^\top(X^\top X + \lambda\mathbf{I})\mathbf{v} = \|X\mathbf{v}\|^2 + \lambda\|\mathbf{v}\|^2 > 0$ for $\mathbf{v} \neq \mathbf{0}$, so the matrix is positive definite, invertible, and J is strictly convex. ■

2.2 MAP interpretation

With prior $\boldsymbol{\beta} \sim N(\mathbf{0}, \tau^2\mathbf{I})$, $-\log p(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{1}{2\sigma^2}\|\mathbf{y} - X\boldsymbol{\beta}\|^2 + \frac{1}{2\tau^2}\|\boldsymbol{\beta}\|^2 + \text{const}$; multiplying by σ^2 gives ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means strong shrinkage.

2.3 Shrinkage in singular directions

Let $X = U\Sigma V^\top$ (thin SVD, singular values s_i). Since $X^\top X + \lambda \mathbf{I} = V(\Sigma^2 + \lambda \mathbf{I})V^\top$,

$$\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i}{s_i^2 + \lambda}(\mathbf{u}_i^\top \mathbf{y})\mathbf{v}_i, \qquad X\hat{\boldsymbol{\beta}}_\lambda = \sum_i \frac{s_i^2}{s_i^2 + \lambda}(\mathbf{u}_i^\top \mathbf{y})\mathbf{u}_i.$$

Least squares fits $\sum_i (\mathbf{u}_i^\top \mathbf{y})\mathbf{u}_i$; ridge multiplies the ith component by $s_i^2/(s_i^2 + \lambda) \in (0, 1)$. Small singular values (directions along which the data barely vary, where least squares amplifies noise by $1/s_i$) are shrunk most. $\lambda \to 0$ recovers least squares; $\lambda \to \infty$ sends coefficients to zero.

2.4 Bias and variance of ridge

Assume $\mathbf{y} = X\boldsymbol{\beta}_0 + \boldsymbol{\varepsilon}$ with $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, and X of full column rank with $X^\top X = VDV^\top$, $D = \mathrm{diag}(d_1, \dots, d_p)$, $d_i > 0$.

Theorem 3.4: ridge trades bias for variance

(i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}_0 = -\lambda(X^\top X + \lambda \mathbf{I})^{-1}\boldsymbol{\beta}_0$; for $\lambda > 0$ the estimator is unbiased iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\mathrm{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2(X^\top X + \lambda \mathbf{I})^{-1} X^\top X (X^\top X + \lambda \mathbf{I})^{-1}$ and

$$\operatorname{tr}\mathrm{Cov}(\hat{\boldsymbol{\beta}}_\lambda) = \sigma^2 \sum_i \frac{d_i}{(d_i + \lambda)^2} < \sigma^2 \sum_i \frac{1}{d_i} = \operatorname{tr}\mathrm{Cov}(\hat{\boldsymbol{\beta}}_0) \qquad (\lambda > 0).$$

(iii) $\lambda \mapsto \|\hat{\boldsymbol{\beta}}_\lambda\|$ is decreasing.

Proof. (i) $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda \mathbf{I})^{-1} X^\top X \boldsymbol{\beta}_0$. From $(X^\top X + \lambda \mathbf{I})^{-1}(X^\top X + \lambda \mathbf{I}) = \mathbf{I}$ we get $(X^\top X + \lambda \mathbf{I})^{-1} X^\top X = \mathbf{I} - \lambda(X^\top X + \lambda \mathbf{I})^{-1}$, which gives the bias. The matrix $(X^\top X + \lambda \mathbf{I})^{-1}$ is invertible, so the bias vanishes iff $\boldsymbol{\beta}_0 = \mathbf{0}$.

(ii) $\hat{\boldsymbol{\beta}}_\lambda = M\mathbf{y}$ with $M = (X^\top X + \lambda \mathbf{I})^{-1} X^\top$, so $\mathrm{Cov} = M(\sigma^2 \mathbf{I})M^\top$. All three matrices are diagonalized by V, giving eigenvalues $\sigma^2 d_i/(d_i + \lambda)^2$; the trace is their sum, and $(d_i + \lambda)^2 > d_i^2$.

(iii) $\|\hat{\boldsymbol{\beta}}_\lambda\|^2 = \sum_i s_i^2 (\mathbf{u}_i^\top \mathbf{y})^2/(s_i^2 + \lambda)^2$ by orthonormality of $\mathbf{v}_i$; each term decreases in λ. ■

3 Other Penalties and the Lasso

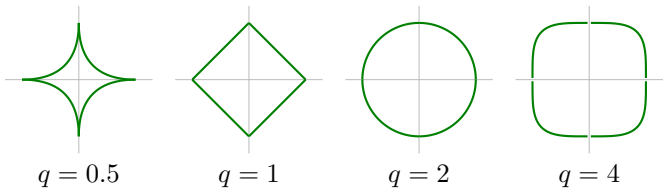
More generally one penalizes $\sum_j |\beta_j|^q$. $q = 2$ is ridge (round level sets, pure shrinkage); $q = 1$ is the lasso, whose diamond-shaped level sets have corners on the axes and produce exact zeros; $0 < q < 1$ is nonconvex.

Theorem 3.5: soft thresholding

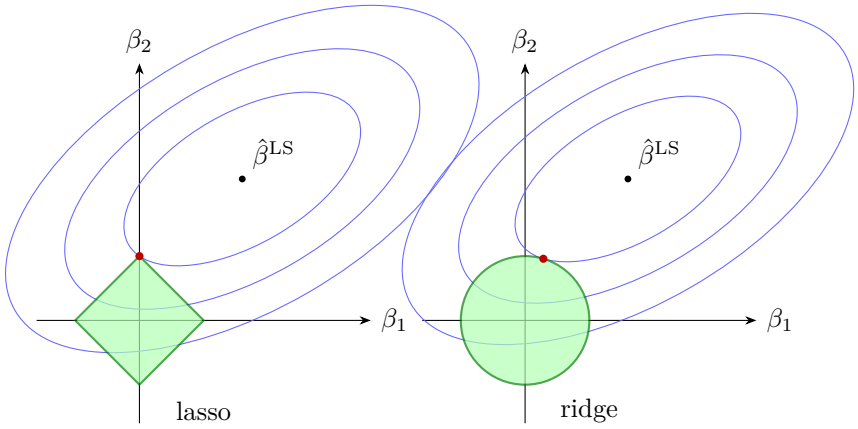
$\arg\min_b \ \frac{1}{2}(b - z)^2 + \lambda|b| = \mathrm{sign}(z)\max(|z| - \lambda, 0)$, whereas $\arg\min_b \frac{1}{2}(b - z)^2 + \frac{\lambda}{2}b^2 = z/(1 + \lambda)$.

Proof. The objective is strictly convex; the optimality condition is $0 \in b - z + \lambda\, \partial|b|$, where $\partial|b| = \{\mathrm{sign}\, b\}$ for $b \neq 0$ and $[-1, 1]$ at 0. If $b > 0$: $b = z - \lambda$, valid iff $z > \lambda$. If $b < 0$: $b = z + \lambda$, valid iff $z < -\lambda$. If $b = 0$: need $z \in \lambda[-1, 1]$, i.e. $|z| \leq \lambda$. The ridge case is the stationarity condition $b - z + \lambda b = 0$. ■

Lasso moves every coefficient by a fixed amount and cuts small ones to zero (sparsity); ridge rescales proportionally and never produces exact zeros.



Unit balls $\sum_j |\beta_j|^q \le 1$ of the penalties: $q = 1$ is the lasso, $q = 2$ ridge.



Constrained least squares: the smallest RSS contour meets the lasso diamond at a corner (a zero coefficient) and the ridge disk at a generic point.

4 Kernel Ridge Regression

Motivation. Nonlinear regression can be done by mapping inputs to a feature space and fitting ridge there. The solution depends on the data only through inner products, so a kernel lets us work with infinitely many features at the cost of an $n \times n$ linear system.

4.1 Feature-space and sample-space forms

Let $\boldsymbol{\phi}(\mathbf{x}) \in \mathbb{R}^m$ be a feature map and, $\Phi = [\boldsymbol{\phi}(\mathbf{x}_1) \cdots \boldsymbol{\phi}(\mathbf{x}_n)] \in \mathbb{R}^{m\times n}$ (columns are samples). Ridge in feature space minimizes $J = \frac{1}{2}\|\mathbf{y} - \Phi^\top \mathbf{w}\|^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2$, so

$$\nabla J = -\Phi\mathbf{y} + (\lambda \mathbf{I}_m + \Phi\Phi^\top)\mathbf{w}, \qquad \hat{\mathbf{w}} = (\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi\mathbf{y}.$$

Theorem 3.6: push-through identity

$(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}\Phi = \Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$. Hence with the Gram matrix $K = \Phi^\top\Phi$, $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j) = \boldsymbol{\phi}(\mathbf{x}_i)^\top \boldsymbol{\phi}(\mathbf{x}_j)$,

$$\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha}, \quad \boldsymbol{\alpha} = (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}, \qquad f(\mathbf{x}) = \boldsymbol{\phi}(\mathbf{x})^\top \hat{\mathbf{w}} = \sum_{i=1}^{n} \alpha_i k(\mathbf{x}_i, \mathbf{x}) = \mathbf{k}_{\mathbf{x}}^\top (K + \lambda \mathbf{I}_n)^{-1}\mathbf{y}.$$

Proof. $\Phi(\lambda \mathbf{I}_n + \Phi^\top\Phi) = \lambda\Phi + \Phi\Phi^\top\Phi = (\lambda \mathbf{I}_m + \Phi\Phi^\top)\Phi$. Multiply on the left by $(\lambda \mathbf{I}_m + \Phi\Phi^\top)^{-1}$ and on the right by $(\lambda \mathbf{I}_n + \Phi^\top\Phi)^{-1}$ (both positive definite). The two identity matrices have different sizes. ∎

Remark 3.7: the kernel trick and cost

$\boldsymbol{\phi}$ appears only through inner products, so only the kernel k is needed. The primal form solves an $m \times m$ system and the dual an $n \times n$ one: the kernel form wins when $m \gg n$ (e.g. the Gaussian kernel, whose feature space is infinite-dimensional). Merely mapping to higher dimensions is feature engineering; computing through k without forming $\boldsymbol{\phi}$ is the kernel trick. An alternative derivation substitutes $\mathbf{w} = \Phi\boldsymbol{\alpha}$ and obtains $K[(K + \lambda\mathbf{I})\boldsymbol{\alpha} - \mathbf{y}] = \mathbf{0}$; when K is singular one may not cancel K, but the displayed $\boldsymbol{\alpha}$ is a solution and all solutions give the same $\hat{\mathbf{w}}$.

4.2 Kernels

- Linear: $k(\mathbf{x}, \mathbf{z}) = \mathbf{x}^\top \mathbf{z}$.
- Quadratic in 2D: $(\mathbf{x}^\top \mathbf{z})^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2$, so $\boldsymbol{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1 x_2)$. In 1D, $(1 + xz)^2$ has $\boldsymbol{\phi}(x) = (1, \sqrt{2}x, x^2)$.
- Gaussian: $k(\mathbf{x}, \mathbf{z}) = \exp(-\|\mathbf{x} - \mathbf{z}\|^2/(2\sigma^2))$; σ is the bandwidth.
- Normalized: $\tilde{k}(\mathbf{x}, \mathbf{z}) = k(\mathbf{x}, \mathbf{z})/\sqrt{k(\mathbf{x}, \mathbf{x})k(\mathbf{z}, \mathbf{z})}$.

Theorem 3.8: Gram matrices are PSD

For any feature-map kernel and any $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{a}^\top K\mathbf{a} = \|\sum_i a_i \boldsymbol{\phi}(\mathbf{x}_i)\|^2 \geq 0$.

Not every similarity function is a valid kernel; positive semidefiniteness of all Gram matrices is the requirement.

4.3 Worked examples

Example 3.9: linear kernel

$\mathbf{x}_1 = (1, 2)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 1)$, $\lambda = 1$: $K = \begin{pmatrix} 5 & 2 \\ 2 & 4 \end{pmatrix}$, $(K + \mathbf{I})^{-1} = \frac{1}{26}\begin{pmatrix} 5 & -2 \\ -2 & 6 \end{pmatrix}$, $\boldsymbol{\alpha} = (1/2, 0)$, $\hat{\mathbf{w}} = \frac{1}{2}\mathbf{x}_1 = (1/2, 1)$. At $\mathbf{x} = (1, 0)$: $\mathbf{k}_\mathbf{x} = (1, 2)$ and $f(\mathbf{x}) = 1/2 = \hat{\mathbf{w}}^\top \mathbf{x}$.

Example 3.10: quadratic kernel

$\mathbf{x}_1 = (1, 1)$, $\mathbf{x}_2 = (2, 0)$, $\mathbf{y} = (3, 2)$, $k = (\mathbf{x}^\top \mathbf{z})^2$, $\lambda = 1$. Then $K_{11} = 4$, $K_{12} = 4$, $K_{22} = 16$,

$$(K + \mathbf{I})^{-1} = \frac{1}{69}\begin{pmatrix} 17 & -4 \\ -4 & 5 \end{pmatrix}, \qquad \boldsymbol{\alpha} = \frac{1}{69}\begin{pmatrix} 43 \\ -2 \end{pmatrix}.$$

At $\mathbf{x}_t = (1, 0)$, $\mathbf{k}_{\mathbf{x}_t} = (1, 4)$ and $f(\mathbf{x}_t) = (43 - 8)/69 = 35/69 \approx 0.507$. Check in feature space: $\boldsymbol{\phi}(\mathbf{x}_1) = (1, 1, \sqrt{2})$, $\boldsymbol{\phi}(\mathbf{x}_2) = (4, 0, 0)$, $\hat{\mathbf{w}} = \Phi\boldsymbol{\alpha} = (35, 43, 43\sqrt{2})/69$ and $\boldsymbol{\phi}(\mathbf{x}_t) = (1, 0, 0)$ give the same $35/69$.

Chapter 4. Classification: Logistic Regression, Softmax and SVM

Logistic regression, multiclass classification, support vector machines

1 Odds, Logit and the Logistic Model

Motivation. For a binary label we need a model of a probability, a number in $[0,1]$, as a function of features. Modelling the log-odds as linear gives the logistic model, whose output is the sigmoid of a linear score.

For a probability p, the odds are $o = p/(1-p)$ ($p = 0.8$ gives odds 4) and

$$\operatorname{logit}(p) = \log \frac{p}{1-p} \in \mathbb{R}, \qquad \sigma(z) = \frac{1}{1+e^{-z}} = \operatorname{logit}^{-1}(z).$$

Definition 4.1: logistic regression

With augmented input $\mathbf{x} = (1, x_1, \ldots, x_d)$ and $y \in \{0,1\}$,

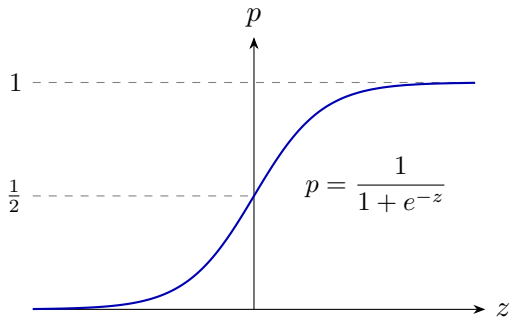
$$\log \frac{P(y=1 \mid \mathbf{x})}{P(y=0 \mid \mathbf{x})} = \mathbf{w}^\top \mathbf{x}, \qquad P(y=1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x}).$$

The score is linear and the probability transform nonlinear. Thresholding the probability at τ gives the hyperplane $\mathbf{w}^\top \mathbf{x} = \log \frac{\tau}{1-\tau}$ (at $\tau = 0.5$: $\mathbf{w}^\top \mathbf{x} = 0$).

Theorem 4.2: sigmoid identities

$\sigma'(z) = \sigma(z)(1-\sigma(z)) \le \frac{1}{4}$, $1 - \sigma(z) = \sigma(-z)$, $\frac{d}{dz}\log(1+e^z) = \sigma(z)$.

Proof. $\sigma'(z) = e^{-z}/(1+e^{-z})^2 = \sigma(z)\cdot\frac{e^{-z}}{1+e^{-z}}$ and $\frac{e^{-z}}{1+e^{-z}} = \frac{1}{e^z+1} = \sigma(-z) = 1-\sigma(z)$. The maximum of $s(1-s)$ on $[0,1]$ is $1/4$ at $s = 1/2$. ■



The logistic function, inverse of the logit $z = \log \frac{p}{1-p}$: log-odds z are mapped to probabilities p.

2 Likelihood, Gradient, Hessian and Convexity

Motivation. Logistic regression has no closed form, so we must optimize numerically. The Hessian is positive semidefinite, so the loss is convex: every local minimum is global and gradient methods are reliable.

For data $(\mathbf{x}_i, y_i)$ with $p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i)$, $p(y_i \mid \mathbf{x}_i, \mathbf{w}) = p_i^{y_i}(1-p_i)^{1-y_i}$ and

$$J(\mathbf{w}) = -\sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right] = \sum_i \left[\log(1+e^{z_i}) - y_i z_i\right], \quad z_i = \mathbf{w}^\top \mathbf{x}_i.$$

Theorem 4.3: logistic loss is convex

$\nabla J = \sum_i (p_i - y_i)\mathbf{x}_i = X^\top(\mathbf{p}-\mathbf{y})$ and $\nabla^2 J = \sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top = X^\top S X$, $S = \mathrm{diag}(p_i(1-p_i))$. Hence J is convex, and strictly convex if X has full column rank.

Proof. $\partial J/\partial z_i = \sigma(z_i) - y_i = p_i - y_i$ by the identities above, and $\partial z_i/\partial \mathbf{w} = \mathbf{x}_i$. Differentiating again, $\nabla p_i = p_i(1-p_i)\mathbf{x}_i$. For any $\mathbf{v}$, $\mathbf{v}^\top X^\top S X \mathbf{v} = \sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 \geq 0$; since $p_i(1-p_i) > 0$, the sum is positive whenever $X\mathbf{v} \neq \mathbf{0}$. ■

Example 4.4: one gradient step

$\mathbf{x} = (1, 2)$, $y = 1$, $\mathbf{w} = \mathbf{0}$: $p = 1/2$, $\nabla J = (p-1)\mathbf{x} = (-1/2, -1)$. With $\eta = 0.1$, $\mathbf{w} = (0.05, 0.1)$, score 0.25, $p \approx 0.562$: the probability of the correct label increased.

3 Labels in $\{\pm 1\}$, Logistic Loss and BCE

Motivation. The same model can be written with labels $\{0, 1\}$ (binary cross-entropy) or $\{\pm 1\}$ (logistic loss of the margin yf). The margin form connects logistic regression with the SVM and the other surrogate losses of this section.

With $Y \in \{\pm 1\}$ and score $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$, $P(Y = +1 \mid \mathbf{x}) = \sigma(f)$ and $P(Y = -1 \mid \mathbf{x}) = \sigma(-f)$, so $-\log P(Y \mid \mathbf{x}) = \ell(Yf)$ with the **logistic loss** $\ell(z) = \log(1 + e^{-z})$. $Yf(\mathbf{x})$ is the *margin*: positive iff the sign of the score is correct.

Theorem 4.5: three names for one objective

Let $P(-1) = (1, 0)$, $P(+1) = (0, 1)$, $\mu(r) = (\sigma(-r), \sigma(r))$, $\tilde{Y} = (1+Y)/2 \in \{0, 1\}$ and $\mathrm{BCE}(s, t) = -[t \log s + (1-t)\log(1-s)]$. For every $(\mathbf{a}, b)$ and every i,

$$D_{\mathrm{KL}}(P(Y_i) \,\|\, \mu(f(\mathbf{x}_i))) = \mathrm{BCE}\left(\sigma(f(\mathbf{x}_i)), \tilde{Y}_i\right) = \ell(Y_i f(\mathbf{x}_i)).$$

So KL minimization, BCE minimization and logistic regression are the same problem.

Proof. For a one-hot first argument, KL $= -\log$(model probability of the true class). If $Y = +1$ this is $-\log \sigma(r) = \log(1+e^{-r}) = \ell(r)$, and BCE with $t = 1$ is $-\log s$. If $Y = -1$ it is $-\log \sigma(-r) = \ell(-r)$, and BCE with $t = 0$ is $-\log(1-s) = -\log \sigma(-r)$. ■

Theorem 4.6: ℓ is convex

$\ell'(z) = \sigma(z) - 1$ and $\ell''(z) = \sigma(z)\sigma(-z) > 0$.

3.1 Separable data

Theorem 4.7: no MLE on separable data

Suppose $Y_i(\mathbf{a}^{\star\top}X_i + b^\star) > 0$ for all i. Then $\inf_{\mathbf{a},b}\sum_i \ell(Y_i(\mathbf{a}^\top X_i + b)) = 0$, the infimum is not attained, and every minimizing sequence is unbounded. With an ℓ_2 penalty $\frac{\lambda}{2}(\|\mathbf{a}\|^2 + b^2)$ a unique minimizer exists and satisfies $\|\mathbf{a}_\lambda\|^2 + b_\lambda^2 \le 2N\log 2/\lambda$.

Proof. Let $t = \min_i Y_i(\mathbf{a}^{\star\top}X_i + b^\star) > 0$ (finitely many positive numbers). Scaling by $C > 0$ and using that ℓ is decreasing, $L(C\mathbf{a}^\star, Cb^\star) \le N\log(1 + e^{-Ct}) \to 0$; and $L \ge 0$. Every term is strictly positive, so $L > 0$ everywhere. If a minimizing sequence were bounded, a convergent subsequence would give a point with $L = 0$. With the penalty: $L_\lambda \ge \frac{\lambda}{2}\|(\mathbf{a}, b)\|^2$ makes the sublevel set $\{L_\lambda \le L_\lambda(\mathbf{0})\}$ compact, so a minimizer exists (Weierstrass); strong convexity gives uniqueness; and $\frac{\lambda}{2}\|(\mathbf{a}_\lambda, b_\lambda)\|^2 \le L_\lambda(\mathbf{0}) = N\ell(0) = N\log 2$. ∎

3.2 Surrogate risk bounds

For $(X, Y) \sim P$ and classifier $f = \operatorname{sign}(g)$, counting $g = 0$ as an error, the event $\{f(X) \ne Y\}$ implies $Yg(X) \le 0$. For a nonnegative, nonincreasing loss ϕ this gives $\phi(Yg) \ge \phi(0)$ on errors and therefore

$$P(f(X) \ne Y) \le \frac{\mathbb{E}[\phi(Yg(X))]}{\phi(0)}.$$

Logistic: $\phi(0) = \log 2$, so $R_{\log} < \varepsilon \Rightarrow P(f(X) = Y) > 1 - \varepsilon/\log 2$. Hinge $\max(0, 1 - u)$ and exponential e^{-u} have $\phi(0) = 1$. Minimizing a convex surrogate thus controls the 0–1 error, which itself cannot be optimized by gradients.

4 Multiclass Classification and Softmax

Motivation. With K classes the sigmoid generalizes to the softmax, which turns K scores into a probability vector. Its cross-entropy loss has the same clean gradient (prediction minus target) as in the binary case.

For Q classes, $W \in \mathbb{R}^{Q\times d}$, scores $s_k = \mathbf{w}_k^\top \mathbf{x}$ and

$$p_k(\mathbf{x}) = \operatorname{softmax}(\mathbf{s})_k = \frac{e^{s_k}}{\sum_r e^{s_r}}, \qquad J(W) = -\sum_{i,k} y_{ik}\log p_k(\mathbf{x}_i).$$

Scores $(2, 1, 0)$ give approximately $(0.665, 0.245, 0.090)$.

Theorem 4.8: softmax calculus

With $\operatorname{LSE}(\mathbf{s}) = \log\sum_r e^{s_r}$ and $\mathbf{p} = \operatorname{softmax}(\mathbf{s})$:

(i) $\operatorname{softmax}(\mathbf{s} + c\mathbf{1}) = \operatorname{softmax}(\mathbf{s})$, $\operatorname{LSE}(\mathbf{s} + c\mathbf{1}) = \operatorname{LSE}(\mathbf{s}) + c$; stable evaluation $\operatorname{LSE}(\mathbf{s}) = M + \log\sum e^{s_r - M}$, $M = \max s_r$.

(ii) $\nabla \operatorname{LSE} = \mathbf{p}$; Jacobian $\partial p_k/\partial s_m = p_k(\delta_{km} - p_m)$, i.e. $\nabla^2 \operatorname{LSE} = \operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top \succeq 0$.

(iii) $\nabla_{\mathbf{s}}(-\log p_y) = \mathbf{p} - \mathbf{e}_y$; for the whole data set $\nabla_W J = (P - Y)^\top X$ (rows: classes).

(iv) $\log\operatorname{softmax}(\mathbf{s}) = \mathbf{s} - \operatorname{LSE}(\mathbf{s})\mathbf{1}$ and log-softmax is idempotent: $\log\operatorname{softmax}(\log\operatorname{softmax}(\mathbf{s})) = \log\operatorname{softmax}(\mathbf{s})$.

(v) $\max_k s_k \le \operatorname{LSE}(\mathbf{s}) \le \max_k s_k + \log Q$.

Proof. (i) e^c cancels.

(ii) Quotient rule; $\mathbf{v}^\top(\operatorname{diag}\mathbf{p} - \mathbf{p}\mathbf{p}^\top)\mathbf{v} = \sum p_k v_k^2 - (\sum p_k v_k)^2$ is the variance of v_K with $K \sim \mathbf{p}$.

(iii) $-\log p_y = -s_y + \mathrm{LSE}(\mathbf{s})$.

(iv) With $\boldsymbol{\ell} = \log\operatorname{softmax}(\mathbf{s})$, $e^{\ell_j} = p_j$ so $\mathrm{LSE}(\boldsymbol{\ell}) = \log\sum p_j = 0$.

(v) $e^M \le \sum e^{s_r} \le Q e^M$. ■

Adding the same vector to all rows of W leaves the softmax unchanged, so W is identifiable only up to such shifts.

5 Support Vector Machines

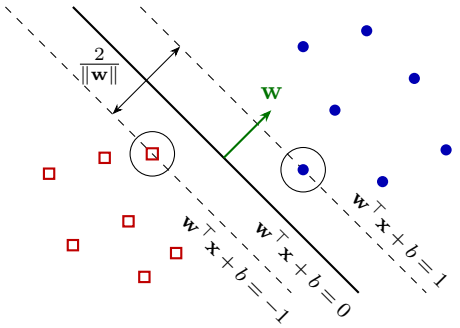
5.1 Geometry and the primal problem

Labels are $y_i \in \{\pm 1\}$, score $f(\mathbf{x}) = \mathbf{w}^\top\mathbf{x} + b$. The distance from $\mathbf{x}_0$ to $\{\mathbf{w}^\top\mathbf{x} + b = 0\}$ is $|\mathbf{w}^\top\mathbf{x}_0 + b|/\|\mathbf{w}\|$ (move along the unit normal: $\mathbf{x}_0 - t\mathbf{w}/\|\mathbf{w}\|$ lies on the plane iff $t = (\mathbf{w}^\top\mathbf{x}_0 + b)/\|\mathbf{w}\|$). Rescaling $(\mathbf{w}, b)$ by $c > 0$ changes nothing, so for separable data normalize the closest functional margin to 1:

Definition 4.9: hard-margin SVM

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top\mathbf{x}_i + b) \ge 1, \ i = 1, \dots, n.$$

The support planes $\mathbf{w}^\top\mathbf{x} + b = \pm 1$ lie at distance $1/\|\mathbf{w}\|$ (the margin) from the boundary; the slab width is $2/\|\mathbf{w}\|$.



Linear SVM: the margin between the hyperplanes $\mathbf{w}^\top\mathbf{x} + b = \pm 1$ is $2/\|\mathbf{w}\|$; the circled points are support vectors.

5.2 Lagrangian, max–min inequality, weak duality

Theorem 4.10: max–min inequality

For any $F : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, $\sup_x \inf_y F(x,y) \le \inf_y \sup_x F(x,y)$. Equality can fail: for $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (rows x, columns y) the left side is 0 and the right side 1.

Proof. For all x, y: $\inf_{y'} F(x,y') \le F(x,y) \le \sup_{x'} F(x',y)$. The left end does not depend on y and the right end not on x; take $\sup_x$ on the left, then $\inf_y$ on the right. ■

With $c_i = 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)$ and $\alpha_i \geq 0$, $L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 + \sum_i \alpha_i c_i$. Since $\sup_{\alpha \geq 0} \alpha c = 0$ if $c \leq 0$ and $+\infty$ if $c > 0$, $\sup_{\boldsymbol{\alpha} \geq \mathbf{0}} L$ equals the objective on feasible points and $+\infty$ elsewhere; hence $p^\star = \inf_{\mathbf{w},b} \sup_{\boldsymbol{\alpha}} L$. The dual is $d^\star = \sup_{\boldsymbol{\alpha}} \inf_{\mathbf{w},b} L$ and the max–min inequality gives **weak duality** $d^\star \leq p^\star$: every dual value is a lower bound on the primal optimum. For the SVM (convex, strictly feasible) strong duality $d^\star = p^\star$ holds.

5.3 The dual problem

Theorem 4.11: SVM dual

$\nabla_{\mathbf{w}} L = \mathbf{0} \Rightarrow \mathbf{w} = \sum_i \alpha_i y_i \mathbf{x}_i$; finiteness of $\inf_b L$ requires $\sum_i \alpha_i y_i = 0$. The dual is

$$\max_{\boldsymbol{\alpha} \geq \mathbf{0}} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \mathbf{x}_i^\top \mathbf{x}_j \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0.$$

Complementary slackness $\alpha_i[1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)] = 0$: $\alpha_i > 0$ only for support vectors, and $b = y_i - \mathbf{w}^\top \mathbf{x}_i$ for any such i.

Proof. b enters L linearly with coefficient $-\sum \alpha_i y_i$; if nonzero, $\inf_b = -\infty$. Substituting $\mathbf{w}$: $\frac{1}{2}\|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 + \sum \alpha_i = \sum \alpha_i - \frac{1}{2}\|\mathbf{w}\|^2$. ■

Example 4.12: two points

$\mathbf{x}_1 = (0,0)$, $y_1 = -1$; $\mathbf{x}_2 = (2,2)$, $y_2 = +1$. The constraint forces $\alpha_1 = \alpha_2 = a$ and $\|\mathbf{x}_2\|^2 = 8$, so the dual is $2a - 4a^2$, maximized at $a = 1/4$. Then $\mathbf{w} = \frac{1}{4}(2,2) = (1/2, 1/2)$, $b = -1$ (from $\mathbf{x}_1$), the boundary is $x^{(1)} + x^{(2)} = 2$, $\|\mathbf{w}\| = 1/\sqrt{2}$, margin $\sqrt{2}$ (slab width $2\sqrt{2}$). Primal value $\frac{1}{2}\|\mathbf{w}\|^2 = 1/4$ equals the dual value $2a - 4a^2 = 1/4$.

Remark 4.13: soft margin and kernels

For non-separable data add slacks: $\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum \xi_i$ s.t. $y_i f(\mathbf{x}_i) \geq 1 - \xi_i$, $\xi_i \geq 0$, equivalently hinge loss $C\sum \max(0, 1 - y_i f(\mathbf{x}_i))$ plus ℓ_2. Its dual is identical except for the box constraint $0 \leq \alpha_i \leq C$. Replacing $\mathbf{x}_i^\top \mathbf{x}_j$ by a kernel gives the kernel SVM with score $\sum_i \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$. Logistic regression fits a likelihood; the SVM maximizes a geometric margin.